

Microsoft Fabric Warehouse

Disaster Recovery & Business Continuity

DBA Technical Reference | Volume 6 of 9

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1. BCDR Overview & Shared Responsibility

Business Continuity and Disaster Recovery (BCDR) in Microsoft Fabric follows a shared responsibility model. Microsoft manages infrastructure resilience; customers are responsible for data-level and application-level recovery planning.

Responsibility	Microsoft Manages	Customer Manages
Infrastructure	Azure datacenter hardware, networking, power	Not applicable
Platform availability	Fabric service availability SLAs	Monitor and respond to outages
OneLake storage replication	Geo-redundant storage (if BCDR enabled)	Enable BCDR toggle in Admin Portal
Warehouse data recovery	Restore-in-place tooling	Initiate restore when needed
Application recovery	Not applicable	Update connection strings, rebind reports
Pipeline recovery	Not applicable	Rerun or reschedule failed pipelines
Permissions post-failover	Not applicable	Reassign workspace roles and SQL permissions

2. Azure Availability Zones

Fabric leverages Azure Availability Zones to protect against datacenter-level failures within a single Azure region. This provides high availability for zone-level outages without any customer action required.

- Availability Zones are physically separate groups of datacenters within the same Azure region
- Fabric resources are automatically distributed across multiple zones — no configuration needed
- During a zone-wide outage, automatic failover occurs to a healthy zone
- Zone failover is transparent — connections are re-routed automatically
- OneLake data engineering (Lakehouse/Warehouse) is zone-resilient by default

INFO: Availability Zones protect against single datacenter failures within a region. For protection against entire region failures, cross-region BCDR replication must be explicitly enabled.

3. OneLake Geo-Redundant Replication

When BCDR is enabled on a Fabric Capacity, OneLake data is asynchronously replicated to the Azure paired region for that capacity. This protects against full regional outages.

Detail	Value
Replication type	Asynchronous (not synchronous — RPO is not zero)
What is replicated	OneLake data: Lakehouses, Warehouses, Notebooks, Spark Job Definitions
What is NOT replicated	Mirrored databases, Power BI semantic models, pipeline definitions
Replication target	Azure paired region for the capacity's primary region
Enablement	Per-capacity toggle in Fabric Admin Portal
Cost	Additional storage charges + CU charges for write replication
Failover initiation	Manual — customer must initiate failover through Microsoft support

WARNING: BCDR replication is asynchronous. There will be a Recovery Point Objective (RPO) gap — data committed in the primary region after the last sync may be lost in a disaster scenario.

4. Enabling BCDR in Fabric Admin Portal

Step-by-step guide to enable BCDR for a Fabric Capacity:

Step 1	Sign in to the Fabric Admin Portal at admin.fabric.microsoft.com
Step 2	Navigate to Capacity settings and select the target capacity
Step 3	In the capacity settings panel, locate the Disaster Recovery toggle
Step 4	Review the cost implications displayed in the dialog
Step 5	Enable the toggle — BCDR replication begins automatically
Step 6	Verify replication status in the Capacity Metrics App
Step 7	Document the paired region for your DR runbook

5. Restore-in-Place (Warehouse Recovery)

Fabric Warehouse supports restore-in-place, which allows you to roll back a Warehouse to a previous known-good state. This is useful for recovering from accidental data corruption, failed deployments, or bad ETL runs.

Restore Characteristic	Details
Restore scope	Entire Warehouse only — cannot restore individual tables
Recovery point	Point-in-time snapshots maintained by Fabric platform
Initiation method	Fabric portal — Warehouse settings > Restore
Downtime required	Yes — the Warehouse is unavailable during the restore operation
Data loss risk	Any data written after the selected restore point will be lost
Use case	Corruption, failed deployment, accidental mass DELETE/UPDATE

WARNING: Restore-in-place restores the ENTIRE Warehouse — you cannot selectively restore individual tables. Always test ETL changes in a development Warehouse before applying to production.

6. Cross-Region DR Strategy

For organizations requiring automated cross-region failover with near-zero RPO, Microsoft recommends maintaining two parallel Fabric Warehouse environments in different Azure regions.

Strategy Component	Implementation Details
Primary region	Production Fabric Capacity in primary Azure region (e.g., East US)
Secondary region	Standby Fabric Capacity in paired Azure region (e.g., West US)
Data parity	Regular ETL pipelines load data to BOTH regions simultaneously
Code parity	Git integration ensures identical schema and code deployed to both
Connection switching	Application connection strings updated to secondary endpoint on failover
Mirrored databases	Must be recreated in secondary region — not replicated automatically
Power BI reports	Reports must be rebound to secondary Warehouse endpoints post-failover

TIP: Use Azure DevOps or GitHub Actions pipelines to deploy schema changes to both primary and secondary Warehouses simultaneously, maintaining code parity automatically.

7. Recovery Procedures Step-by-Step

7.1 Zone-Level Outage Recovery

- Zone failover is automatic — no customer action required
- Monitor Fabric Service Health dashboard for outage notifications
- Connections are automatically rerouted — applications should reconnect with retry logic

7.2 Regional Outage Recovery (BCDR Enabled)

Step 1: Monitor Microsoft Azure Service Health for regional outage confirmation

Step 2: Contact Microsoft Support to initiate Fabric regional failover

Step 3: After failover, navigate to the secondary region Fabric workspace

Step 4: Create new Warehouse items with the same names as the primary region

Step 5: Restore OneLake data using the scripted recovery approach provided by Microsoft

Step 6: Reassign workspace roles — roles are not replicated automatically

Step 7: Update SQL endpoint permissions (GRANT statements) on recovered Warehouses

Step 8: Recreate mirrored databases — these are not replicated by BCDR

Step 9: Rebind Power BI reports to the secondary Warehouse SQL endpoints

Step 10: Update application and pipeline connection strings to secondary endpoints

Step 11: Re-enable and reschedule any paused pipelines and data flows

8. Git Integration for DR

Git integration is one of the most valuable tools for accelerating Warehouse recovery. By keeping all schema objects, stored procedures, and configurations in version control, recovery becomes a scripted, repeatable process.

Git Integration Benefit	DR Impact
Schema in source control	Recreate all tables, views, and procedures from scripts in minutes
Automated deployment pipelines	Deploy schema to secondary region in a single pipeline run
Environment consistency	Ensure primary and secondary are always in sync
Rollback capability	Roll back a bad deployment without needing restore-in-place
Change history	Identify exactly what changed before a corruption event

9. RTO & RPO Planning

DR Scenario	Typical RTO	Typical RPO	Cost Level
Zone-level outage (automatic)	Minutes	Near-zero	Included
Restore-in-place	30 min - 2 hours	Hours to days (last snapshot)	Low
BCDR regional failover (OneLake)	Hours (manual steps)	Minutes to hours (async lag)	Medium
Active-active dual region	Minutes (connection switch)	Near-zero (dual writes)	High

INFO: RTO = Recovery Time Objective (how long to restore service). RPO = Recovery Point Objective (how much data loss is acceptable). Define these with business stakeholders before selecting a DR strategy.

10. BCDR Best Practices & Checklist

Area	Checklist Item	Done?
Planning	Define RTO and RPO requirements with business owners	☐
Planning	Document all Fabric workspaces and their region assignments	☐
Enablement	Enable BCDR toggle for all production capacities	☐
Enablement	Verify paired region is documented in DR runbook	☐
Code	All schema objects stored in Git repository	☐
Code	Automated deployment pipeline to secondary region exists	☐
Testing	Run DR drill at least once per quarter	☐
Testing	Test restore-in-place on non-production Warehouse	☐
Documentation	DR runbook includes all manual post-failover steps	☐
Documentation	Connection string failover procedure documented for each application	☐
Monitoring	Azure Service Health alerts configured for Fabric	☐
Monitoring	Capacity Metrics App monitored for replication health	☐